

# Net-zero CO<sub>2</sub> by 2070 in the SCI ensemble: what a 2010–2025 hindcast can and cannot tell us

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## Abstract

Of the 1,564 pathways in the Scenario Compass Initiative (SCI) 2025 ensemble, 497 reach net-zero CO<sub>2</sub> by 2070, a naive frequency  $P(\text{NZ2070}) \approx 32\%$ . We ask what the observed 2010–2025 record establishes about that figure. First, each pathway is a *conditional* forecast, so the count is a sample frequency, not a probability: it moves between 16% and 57% with the chosen deadline and between 24% and 39% with the team-weighting, neither a climate quantity. Second, scored against observations for six variables, the ensemble sits below observed 2025 emissions (+2,582 Mt CO<sub>2</sub>/yr, family-weighted); the shortfall concentrates in net-zero pathways; a trivial extrapolation beats it on the on-trend variables; and the world is in an *addition* regime, solar roughly double the ensemble median while coal set a record. Filtering for accuracy moves the share to about 20% under sensible criteria—a sensitivity, not a revised probability. Independently, Wright’s-law cost projections nearly coincide for net-zero and non-net-zero pathways, locating the binding constraint in fossil substitution rather than in the cost of clean technology.

## Keywords

hindcast, scenario ensemble, net zero, conditional forecast, integrated assessment models

## 1 Introduction

The SCI 2025 ensemble assembles 1,564 IAM pathways from 65 models across 41 research projects. Counting the 497 that reach net-zero CO<sub>2</sub> by 2070 gives  $P(\text{NZ2070}) \approx 32\%$ , which treats all pathways as equally credible draws from the distribution of futures. We test that interpretation along two lines—what the number is (Sec. 3), and whether the ensemble behaves like a usable forecast when scored on the recent past (Sec. 4)—then ask what a 15-year record can and cannot settle about a 45-year question (Sec. 5). We do not propose a corrected probability; we report where the evidence is firm and where it is not.

## 2 Data and method

We compare SCI-2025 (v1.0, global) projections at  $t \in \{2010, 2015, 2020, 2025\}$  with observed values for six variables: CO<sub>2</sub> emissions [1], coal primary energy [2], solar PV and wind capacity [3], nuclear capacity (IAEA PRIS), and GDP (PPP, World Bank). For pathway  $j$ , variable  $v$ , year  $t$ , the forecast error is  $\varepsilon_{j,v,t} = y_{v,t} - \hat{y}_{j,v,t}$  ( $\varepsilon > 0$ : the pathway projected too little), summarised by mean error ME (bias), MAE (magnitude), and RMSE. A pathway is *net-zero-by-2070* if its SCI year-of-net-zero (CO<sub>2</sub>) is  $\leq 2070$ . Three further statistics: *skill* =  $|\varepsilon_{\text{ens}}|/|\varepsilon_{\text{rule}}|$  against a naive 2010–2015 random-walk or linear-trend rule ( $> 1$  means the rule wins); *calibration*, the percentile (PIT)

at which the observed value falls in the ensemble cloud, near the median if calibrated; and a *credibility filter* that retains the lowest-MAE quartile on the selected variables. Because the ensemble is a convenience sample—five families contribute about two-thirds of the runs—we weight each pathway by  $1/n_f$  ( $n_f$  = runs from its family) and check robustness against scenario- and project-weighting. This is a *hindcast* (a backtest on a window each model could see), not an out-of-sample test: all pathways post-date 2017.

## 3 The naive count is a sample frequency, not a probability

A pathway states  $\hat{y} = f(P)$ : an outcome *given* an assumed policy/technology path  $P$ . Counting conditional “if–then” statements does not yield an unconditional probability, and the count is elastic to choices unrelated to the climate. Two such degrees of freedom each move it by roughly half its range (Table 1): the net-zero *deadline*, and the *weighting* that sets how many runs each team contributes. A quantity this sensitive to arbitrary conventions is not estimating a probability; we therefore test the ensemble as a forecast instead.

**Table 1: The net-zero share moves with two non-climate choices.**

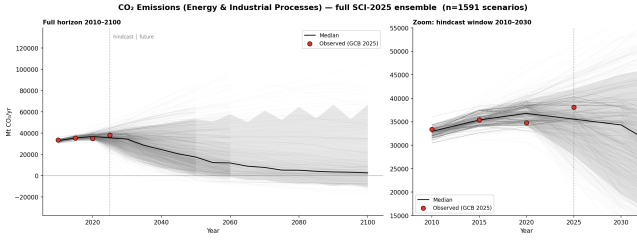
| Deadline     | 2060         | 2070       | 2080        | ever ( $\leq 2100$ ) |
|--------------|--------------|------------|-------------|----------------------|
| Share        | 16%          | 31%        | 43%         | 57%                  |
| Weighting    | per scenario | per family | per project |                      |
| Share (2070) | 31%          | 24%        | 39%         |                      |

## 4 Hindcast diagnosis

*The 2025 undershoot is structural.* Family-weighted CO<sub>2</sub> errors are +405 (2010), +15 (2015), −1,572 (2020), +2,582 Mt/yr (2025). The 2020 overshoot is mostly the COVID dip: replacing 2020 by a shock-free interpolation removes about three-quarters of it. The 2025 shortfall is not an artefact—observed emissions lie on the pre-COVID trend while the ensemble assumed an earlier peak-and-decline (Fig. 1)—and it survives detrending.

*The shortfall concentrates in net-zero pathways.* Mean error by outcome is +650 (net-zero) versus +216 Mt/yr (non-net-zero), family-weighted. The non-net-zero *level* is weighting-dependent (near zero under scenario-weighting), so we read the *gap*, robust at +434/+837/+1,281 under family/scenario/project weighting. Typical magnitudes are close (MAE  $\approx 6\%$  vs  $5\%$ ): this is a directional bias, not greater imprecision, and it is partly definitional—reaching net-zero by 2070 forces an early downturn the record did not show.

*No skill, and biased low as a distribution.* Forecasting 2025 from 2010–2015, a naive rule beats the ensemble on 4 of 6 variables



**Figure 1: The SCI CO<sub>2</sub> ensemble (grey) and the four observed points (black). Observed 2025 emissions sit above the ensemble central tendency.**

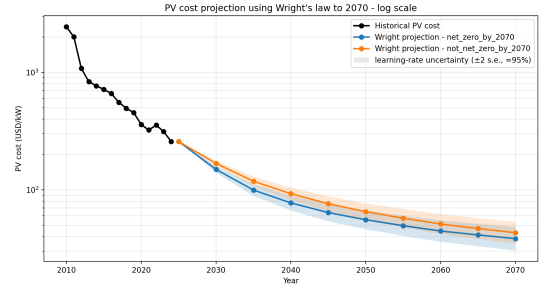
(skill: CO<sub>2</sub> 2.0, coal 4.7, GDP 3.0, nuclear 9.9), losing only where deployment accelerated (solar 0.7, wind 0.6). This restates the bias rather than adding independent evidence—the rule wins because reality stayed on trend while the ensemble bet on a turn—and the nuclear value is hollow (capacity is flat, so “no change” wins by construction). As a distribution the ensemble is biased low: observed 2025 falls at the 1st percentile for GDP, 90th for solar, and 75th–79th for CO<sub>2</sub>, coal and wind. With one point per variable this is a low-bias diagnostic, not a formal calibration test, but the direction is clear: reality is off-centre, so the cloud is not a distribution to read  $P(\text{NZ2070})$  from.

*Addition, not substitution.* Models underestimate fossil and renewable deployment at once: observed solar in 2025 (2,392 GW) is about double the ensemble median (1,169 GW), yet coal set a record (164 EJ). Within the ensemble, coal and solar errors are anti-correlated ( $\rho = -0.28$ ; stronger for net-zero pathways), so the ensemble embeds a substitution logic, but the world adds renewables on top of persistent fossils rather than replacing them. The variables that separate net-zero from non-net-zero errors are exactly coal, CO<sub>2</sub> and solar (confirmed by an  $L_1$ -penalised classifier); their signs disagree (net-zero worse on coal/CO<sub>2</sub>, better on solar), so credibility must be judged jointly—CO<sub>2</sub> alone is misleading, its GDP and intensity errors partly cancelling.

*Filtering: a sensitivity, not a revision.* Keeping the most accurate quartile and recomputing the share gives about 20% (CO<sub>2</sub>), 22% (joint coal+CO<sub>2</sub>+solar), and 48% (solar-only) against the ~32% baseline. The robust feature is the *dependence* on the conditioning variable—the conditional-forecast point of Sec. 3: no single unconditional share exists. We report ~20% under sensible criteria; the 48% arises only from filtering on solar, the variable every model misses.

## 5 What a 15-year record cannot settle, and what it can

A backtest over 15 years cannot resolve a 45-year question: for a drifting random walk the forecast-error variance grows as  $\sigma^2(\tau + \tau^2/m)$  in the horizon  $\tau$ . A pathway flat to ~2030 then falling steeply to net-zero is indistinguishable from a non-net-zero pathway over 2010–2025, so filtering cannot push below the late-mover share (~20% as a practical, not proven, floor). One quantity does carry skill: technology cost. Solar PV fell from ~2,440 to ~250 USD/kW (2010–2024); a Wright’s-law fit (cost  $\propto$  cumulative-capacity<sup>- $b$</sup> , projected along each group’s capacity path) gives ~30–40 USD/kW by



**Figure 2: Wright’s-law PV cost projection to 2070; net-zero and non-net-zero trajectories nearly coincide (central lines, not uncertainty bands).**

2070, with net-zero and non-net-zero projections nearly coincident (Fig. 2). Cost is therefore not the binding constraint—clean technology outperforms the scenarios whether or not the world reaches net-zero. The two halves reconcile: the ensemble is too optimistic on *emissions* yet too pessimistic on *cost*, because the world both deploys renewables faster than any model and keeps its fossils. The lock on net-zero is substitution, not the price of clean energy.

## 6 Conclusion

None of this forecloses net-zero by 2070; it relocates the constraint. The naive 32% is a sample frequency, not a probability; its probabilistic reading does not survive the record; and credibility filtering gives a sensitivity centred near 20% that a 15-year window cannot sharpen. The decisive variable is substitution—policy and system inertia retiring fossils—not the cost of clean technology. Three caveats bound these claims: 2020–2025 contains the COVID shock, which no pathway was built to capture; all pathways post-date 2017, so this is a hindcast, not a true out-of-sample test (an AR5-vintage test, ~2014 forecasting 2025, would provide one); and the calibration evidence rests on one observed point per variable. Within those bounds, the systematic, cross-model optimism about fossil phase-out is a structural feature of the ensemble rather than a transient one.

## References

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